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Text Analytics Course

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Exploring Mental Health Conditions Through Linguistic Patterns in Online Texts

Project Report

Group 2

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Abstract

This study evaluates Natural Language Processing (NLP) methods for the multi-class classification of mental health conditions from written text. Using the Kanakmi Mental Disorders Dataset, which comprises more than 580,000 forum posts across six diagnostic categories, three supervised learning approaches are compared: Support Vector Machines, XGBoost, and Multi-Layer Perceptron networks. The employed text representations combine contextual embeddings with linguistically motivated features designed to capture lexical, emotional, and stylistic properties of informal online language.

Experimental results indicate that the Multi-Layer Perceptron achieves the highest test accuracy (0.69), followed by SVM (0.68) and XGBoost (0.66). Performance is observed to vary across diagnostic categories, with superior results recorded for Anxiety and Schizophrenia compared to Bipolar disorder. Furthermore, interpretability analysis using LIME and SHAP reveals that neural models employ a hybrid decision process, integrating semantic context with specific linguistic markers such as emotional vocabulary and profanity-related patterns.

Collectively, these findings suggest that the integration of hybrid representations with interpretable models can enhance the reliability and transparency of NLP systems for mental health text classification.

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Introduction and State of the Art

Mental health disorders affect over one billion people worldwide and represent a major global health and economic challenge. Limited public investment, unequal access to care, and a shortage of trained professionals highlight the need for complementary approaches that support early identification and intervention [9]. In this context, written language offers a valuable source of information, as individuals often express emotional and psychological states through personal texts.

Recent advances in Natural Language Processing (NLP) have demonstrated the potential of computational methods for mental health assessment. Early research established that individuals with mental health conditions exhibit distinctive linguistic patterns in their social media communications, with measurable differences observed across multiple disorders including ADHD, bipolar disorder, depression, PTSD, schizophrenia, and seasonal affective disorder [2]. The field has evolved from traditional machine learning approaches using lexical features, TF-IDF representations, and psycholinguistic dictionaries like LIWC, to sophisticated neural architectures including transformer-based models such as BERT that achieve state-of-the-art performance in detection tasks [10]. Beyond detection, NLP applications now extend to mental health monitoring, treatment support, and intervention delivery, though significant challenges persist regarding clinical validation, ethical considerations, and the gap between methodologically advanced techniques and their practical applicability in clinical settings, since clinical applications require understanding which language features contribute to mental health assessments [5].

In parallel, recent work has also highlighted the importance of incorporating social determinants of health into NLP-based mental health research, recognizing that factors such as socioeconomic status, access to care, and demographic characteristics significantly influence mental health outcomes and should be considered in model development and evaluation [8]. While initial research demonstrated the possibility of differentiating among various mental health conditions through language analysis [2], much of the subsequent research has focused on binary classification tasks, typically detecting the presence or absence of a single condition. Questions about the comparative classification of multiple mental health disorders and generalizability across different populations and data sources remain areas requiring further investigation.

This study investigates whether linguistic analysis of written content can reveal patterns associated with different mental health conditions. The analysis is based on the Kanakmi Mental Disorders Dataset, which contains approximately 581,000 English texts collected from online mental health–related forums. Each record consists of a textual post and a corresponding label encoding one of these six categories: [0: BPD, 1: Bipolar, 2: Depression, 3: Anxiety, 4: Schizophrenia, 5: MentalIllness]. By addressing multi-class classification across six distinct conditions, this work extends beyond binary detection approaches and examines the comparative discriminative power of different feature representations and model architectures. The project explores text-based feature representations and evaluates multiple supervised learning models, including Support Vector Machines, XGBoost, and neural networks, complemented by explainability methods to support the interpretation of classification outcomes.

1 Data Preparation and Preprocessing

The data preparation and preprocessing workflow transforms raw text into a structured, multidimensional numerical representation while minimizing linguistic noise. The pipeline is organized into four key stages: normalization (using the spaCy library), feature extraction, semantic vectorization, scaling and encoding.

1.1 Data Understanding and Visualization

An initial exploratory analysis was conducted to examine the main characteristics of the dataset prior to modeling. The dataset comprises a large and heterogeneous collection of forum-style texts related to mental health, labeled according to six diagnostic categories. An inspection of class frequencies (Figure 1.1) revealed a strong imbalance among categories, with BPD and anxiety-related texts being substantially more frequent than others.

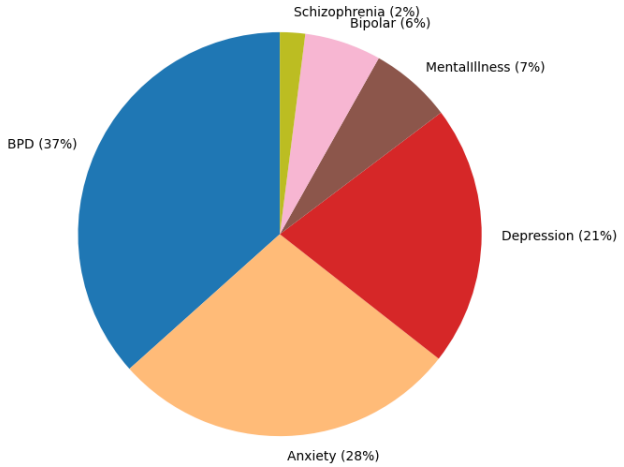


Figure 1.1: Class distribution.

The analysis of text length distribution (Figure 1.2) showed a high degree of variability, ranging from extremely short entries (minimum length of 1 character) to very long narratives exceeding 30,000 characters. This distribution suggests the coexistence of low-information samples and potential outliers, both of which may negatively affect feature extraction and model performance.

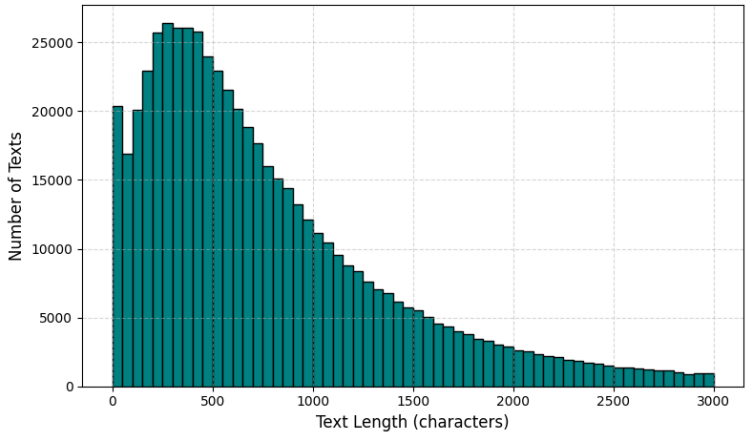


Figure 1.2: Distribution of text lengths within the 0–3000 character range, highlighting the prevalence of short texts and the presence of outliers in the upper tail.

1.2 Text Cleaning

As a first step, non-English texts were removed using *LangDetect* library in order to ensure linguistic consistency across samples. Texts shorter than 200 characters were filtered out to ensure adequate semantic content, resulting in the removal of 68,186 texts (12.00% of the dataset). Texts that were empty or composed exclusively of punctuation, numbers, or symbols were also removed, as they provide limited semantic information.

Additional cleaning steps were performed to address common sources of noise in online text, duplicated entries, and residual markup such as HTML or XML tags (removed using the *BeautifulSoup* library). Text normalization techniques were applied to handle encoding inconsistencies, normalize Unicode characters, and standardize whitespace and line terminators.

Furthermore, an exploratory analysis was conducted to assess the presence and role of emojis within the dataset. The analysis revealed that emoji usage was extremely sparse, with only 3.29% (19,150 texts) con-

taining emojis. While usage varied slightly across mental health conditions, with bipolar disorder and anxiety showing the highest presence (4.0-4.2%) and schizophrenia and depression the lowest (2.3%), these differences were minimal (range: 2.3-4.2%) (Figure 1.3). Given the overall low prevalence and limited variation across labels, emojis were removed to simplify the feature space.

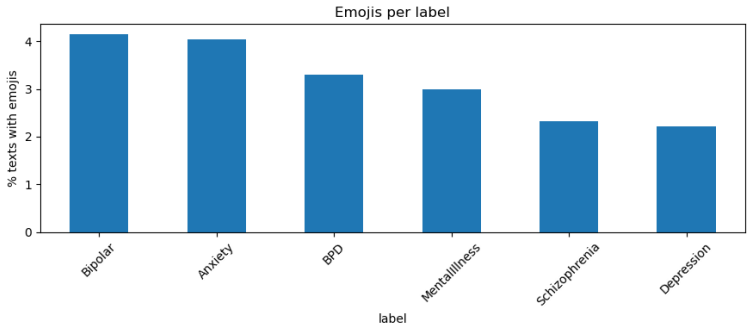


Figure 1.3: Distribution of emoji usage across mental health categories, expressed as the percentage of texts containing emojis.

To improve dataset reliability, outliers in text length were identified using an interquartile range (IQR). A conservative threshold multiplier ($k = 4$) was selected to define the upper bound (approximately 4,443 tokens), ensuring that only extreme anomalies were discarded while preserving the natural variability of the narratives (Figure 1.4). At the end of this process 6393 records were removed from the dataset.

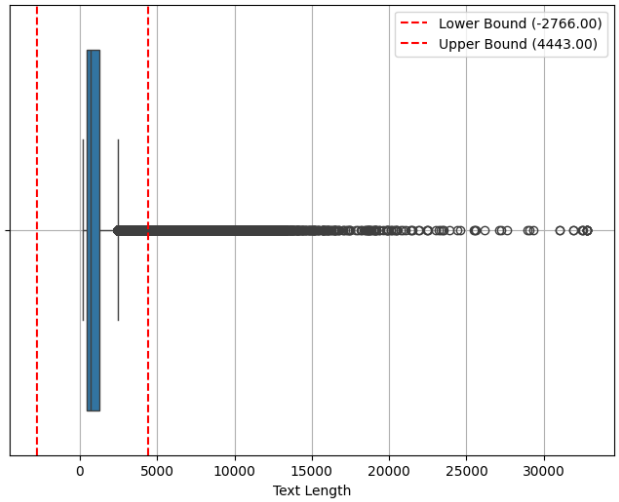


Figure 1.4: Boxplot analysis of text length distribution.

Finally, the generic `MentalIllness` category was excluded to focus the analysis on more specific diagnostic classes.

1.3 Handling Class Imbalance and Text Normalization

To address the significant class imbalance observed in the original dataset and to optimize computational resources for model training, a random undersampling strategy was implemented. A maximum threshold of 8,000 samples per class was established. For categories exceeding this limit, a random subset of 8,000 instances was extracted using a fixed seed (`random_state=42`) to ensure reproducibility. Categories with fewer than 8,000 instances were retained in their entirety.

This procedure resulted in a balanced corpus of 39,976 records, which was subsequently shuffled to eliminate any ordering bias. This preprocessing step ensures that the model is not biased toward majority classes

while maintaining a sufficient volume of data for effective learning. The final distribution of the classes is summarized in Table 1.1.

Label	Original Count	Final Count (Sampled)
Label 0	180,055	8,000
Label 1	27,893	8,000
Label 2	97,501	8,000
Label 3	135,264	8,000
Label 4	7,976	7,976
Total	**448,689**	**39,976**

Table 1.1: Comparison of class distribution before and after the random undersampling process.

Text normalization was performed using token-level processing with the `spaCy` library. To improve efficiency, non-essential components (parser, NER, tagger) were disabled, and the process was parallelized across multiple cores. The preprocessing pipeline involved several sequential steps to standardize the textual data: posts were first tokenized and converted to lowercase to ensure uniform representation. Subsequently, lemmatization was applied to reduce tokens to their base forms, thereby minimizing inflectional variation. To further refine the dataset, noise removal was conducted by filtering out stop words and punctuation marks, ensuring that only content-bearing terms were retained. The resulting normalized text was reconstructed as a sequence of lemmatized tokens, providing a clean and structured input for the subsequent stages.

1.4 Feature Engineering

Feature engineering integrates structural, lexical, syntactic, and semantic information in order to capture complementary aspects of mental health–related language. Since these features are derived from individual texts and do not depend on global frequency information, they were safely computed prior to the train/test split without risk of data leakage.

At the lexical level, text length after normalization was computed to represent text size. As shown in the distribution of cleaned text length (Figure 1.5), the data is characterized by a marked positive skew: the majority of posts are concise (under 100 tokens), while a "long tail" represents a subset of more extensive and detailed narratives.

Lexical diversity was quantified through the Standardized Type–Token Ratio (STTR), a length-independent measure based on fixed-size token windows that provides a robust estimate of vocabulary richness across texts of varying length. The distribution of STTR scores (Figure 1.6) shows a concentration of high values, indicating a consistently rich vocabulary across the dataset.

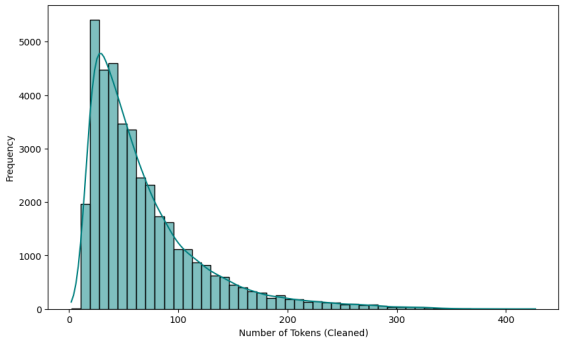


Figure 1.5: Distribution of text length (tokens). The graph shows a marked positive skew with a high concentration of short posts.

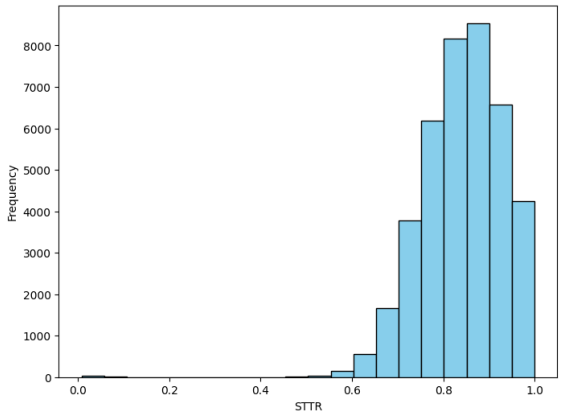


Figure 1.6: Distribution of STTR scores.

Syntactic characteristics were incorporated by computing the average sentence length, measured as the

mean number of tokens per sentence within each text. This feature reflects sentence complexity and narrative organization, which may vary across different forms of personal and emotionally charged writing. As illustrated in the distribution of average sentence length (Figure 1.7), the majority of texts exhibit moderate structural complexity, while the right-skewed tail represents more elaborate or detailed narrative styles.

Beyond sentence-level syntax, discourse structure was further modeled through an average sentence coherence score, obtained by computing the mean cosine similarity between embeddings of consecutive sentences. The distribution of coherence scores (Figure 1.8), reveals a high degree of semantic consistency across the dataset, with most narratives maintaining a logical flow, while lower values identify more fragmented or disjointed discourse patterns.

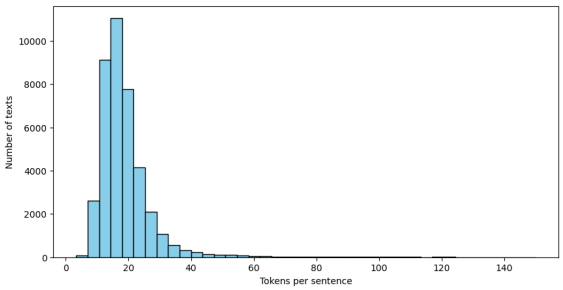


Figure 1.7: Distribution of average sentence length.

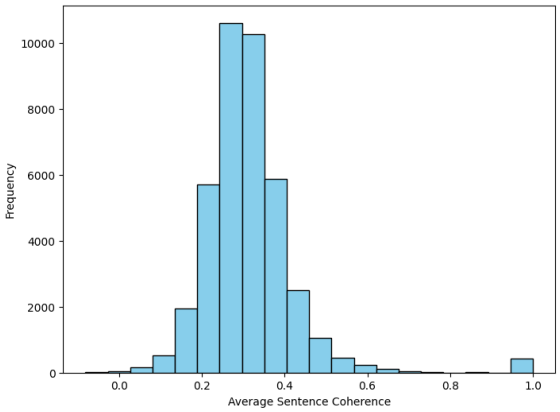


Figure 1.8: Distribution of average sentence coherence scores.

In addition to structural and syntactic features, emotional content was modeled by assigning each text a dominant emotion label predicted by a RoBERTa-based GoEmotions classifier. As shown in the bar plot (Figure 1.9) the most prevalent emotions identified are sadness, disappointment, curiosity, fear, and nervousness. The high frequency of these labels provides a nuanced pragmatic signal: while sadness and nervousness align with the expected affective state of mental health narratives, the presence of curiosity suggests an active engagement or search for understanding within the discourse.

Furthermore, each text was associated with a toxicity probability score, capturing the presence of potentially toxic or aggressive language using a BERT-based transformer from the Detoxify library. To address the heavy concentration of raw scores near zero (typical of non-toxic datasets), a logit transformation was applied to improve numerical dispersion and enhance its usability in downstream models (Figure 1.10). The distribution of the transformed scores is characterized by a high frequency of strongly negative values, confirming that the vast majority of the corpus consists of "clean" or non-aggressive content.

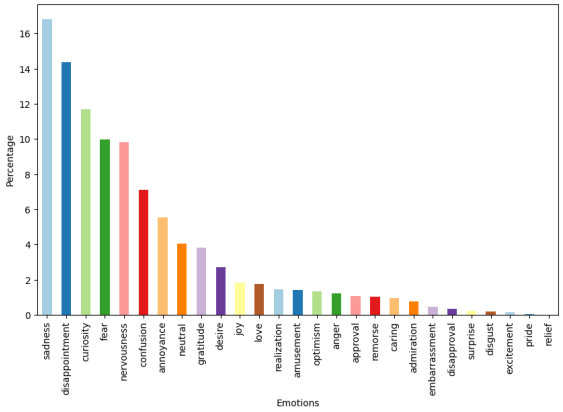


Figure 1.9: Distribution of predicted emotions.

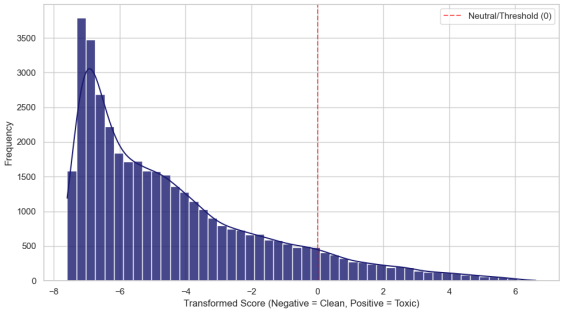


Figure 1.10: Distribution of transformed toxicity scores (logit).

Overall emotional tone was summarized through sentiment polarity classification (negative: 81.1%, neutral: 12.0%, or positive: 7.0%). This analysis was performed using a RoBERTa-base model specifically fine-tuned for sentiment analysis on social media text (`cardiffnlp/twitter-roberta-base-sentiment`). The overwhelming prevalence of negative sentiment is consistent with the nature of the dataset, which focuses on narratives related to psychological distress.

To further explore the affective depth of the corpus, we analyzed the intersection between sentiment polarity and the previously identified emotions (Figure 1.11). As expected, categories such as anger, disgust, and annoyance show a near-total overlap with negative polarity.

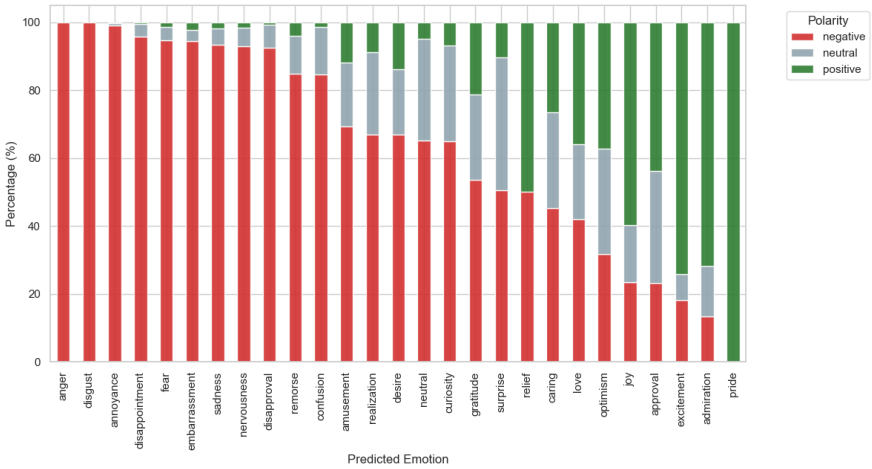


Figure 1.11: Sentiment polarity distribution by predicted emotion.

Following the train-test split, the rare words percentage distribution was computed to capture the proportion of infrequent tokens within each document. To strictly prevent data leakage, the rarity threshold was defined exclusively using word frequencies from the training set: any word appearing five times or fewer across the training corpus was classified as rare. As shown in the histogram (Figure 1.12), most of the texts contain a very low proportion of rare words, with the distribution peaking sharply at 5%, which indicates that the narratives primarily rely on common core vocabulary.

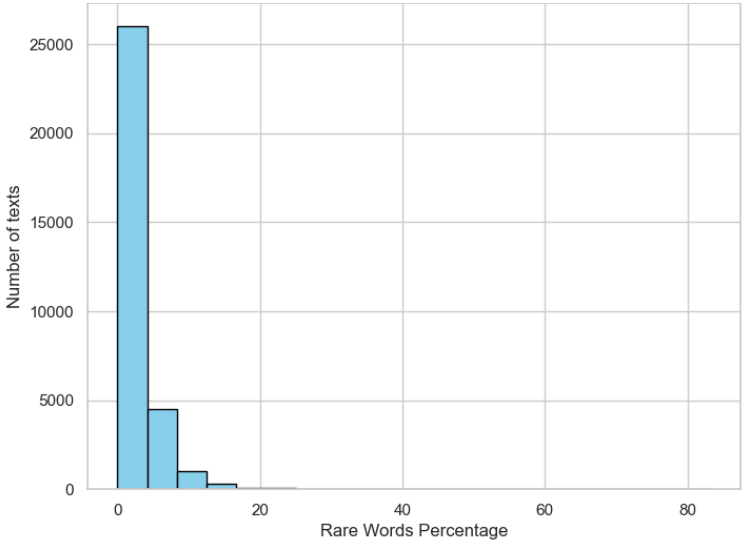


Figure 1.12: Distribution of rare words percentage.

Finally, the engineered features were integrated with deep semantic representations to create a compre-

hensive input for the classification models. DistilBERT was utilized to extract 768-dimensional contextual embeddings from the `[CLS]` token, providing a global summary of each narrative's latent meaning. Categorical variables, including predicted emotions and sentiment polarity, were converted into numerical format through one-hot encoding.

2 Model Implementation

This chapter details the implementation of three distinct architectures for the multiclass classification of mental health textual data: a Linear SVM, an XGBoost ensemble, and a Neural Network (MLP). The objective is to evaluate the performance of different mathematical frameworks in processing a high-dimensional feature set that integrates DistilBERT embeddings with engineered structural and affective metrics.

Each model was optimized using cross-validation and assessed on an independent test set to ensure robust generalization. Performance is measured across five clinical categories using Accuracy, weighted F1-score, and Area Under the ROC Curve (AUC) to provide a comprehensive comparison of their predictive capabilities.

2.1 Support Vector Machine (SVM)

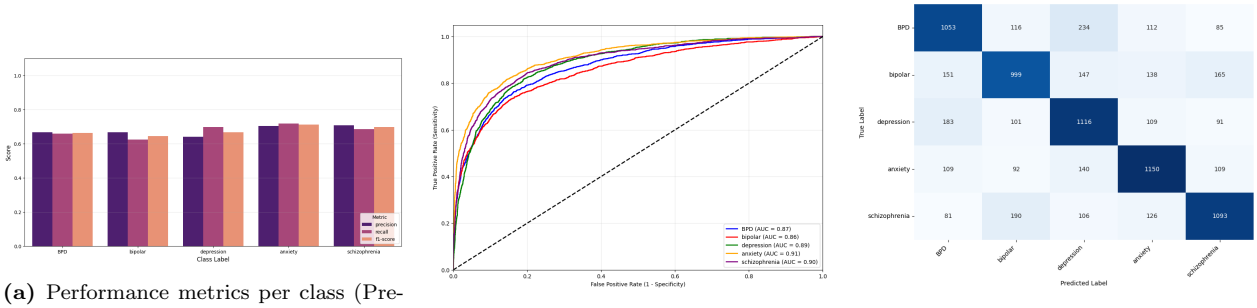
Following the feature engineering phase, a Support Vector Machine (SVM) with a linear kernel was implemented as the primary classification model. Prior to training, data normalization was performed using a `StandardScaler` (Z-score normalization) to ensure uniform feature scaling. To optimize computational efficiency and memory usage given the 805-dimensional input vector, the data was cast to 32-bit floating-point precision (`float32`).

To determine the optimal configuration, a Grid Search with stratified 5-fold cross-validation was conducted. This process identified a regularization parameter of $C = 0.001$ as the best fit, yielding a cross-validation weighted F1-score of 0.668.

The final evaluation on the test set resulted in an overall accuracy and a weighted F1-score of 0.68. As shown in the performance metrics plot (Figure 2.1a), precision and recall remained balanced across all five categories, with *Anxiety* and *Schizophrenia* demonstrating the strongest predictive performance (F1-scores of 0.71 and 0.70, respectively).

The model's robust discriminative power is further highlighted by the Multiclass ROC curves (Figure 2.1b), where the Area Under the Curve (AUC) values range from 0.86 to 0.91. These high AUC values confirm that the classifier maintains a high sensitivity and specificity across all mental health conditions. The confusion matrix (Figure 2.1c) confirms this reliability, illustrating a high rate of correct predictions across all categories, as indicated by the dense diagonal.

Furthermore, the high degree of consistency between the training cross-validation score (0.668), validation (0.67), and test (0.68) results indicates that the model effectively avoided overfitting, successfully demonstrating its power to generalize to unseen, real-world discourse.



(a) Performance metrics per class (Precision, Recall, F1-Score).

(b) Multiclass ROC curves.

(c) Confusion matrix.

Figure 2.1: Linear SVM Evaluation

2.2 eXtreme Gradient Boosting (XGBoost)

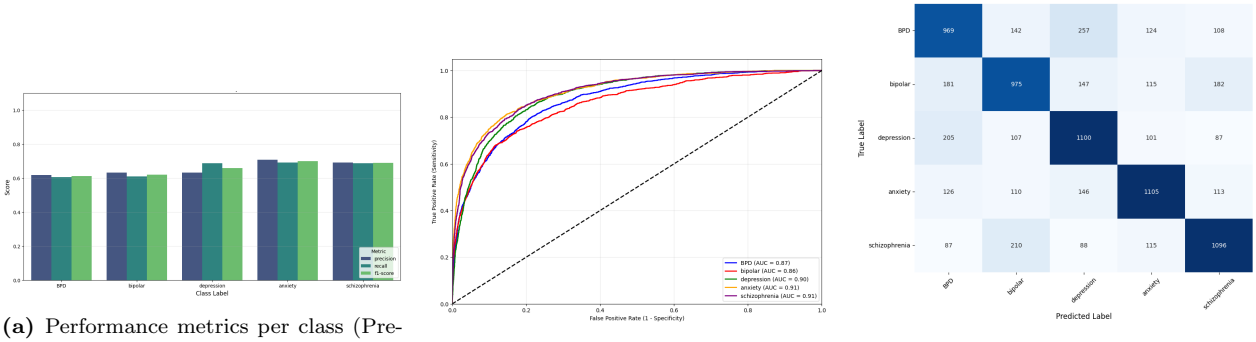
An XGBoost classifier was implemented to capture complex non-linear feature interactions. Since tree-based models are inherently robust to varying scales, explicit normalization was omitted.

To optimize the model's architecture, a Bayesian optimization process was conducted using the Optuna framework. The optimization spanned 100 trials, maximizing the Macro F1-Score within a Stratified 5-Fold Cross-Validation loop. From a computational perspective, the training was hardware-accelerated using NVIDIA GPUs with histogram-based split finding (`tree_method='hist'`) and minimized the Multiclass Logarithmic Loss (mlogloss) to refine probabilistic confidence.

The optimization identified a robust configuration characterized by a conservative learning rate of 0.08, a shallow tree depth of 4 (preventing overfitting on noise), and a high number of estimators (1,569). Crucially, a significant L2 regularization ($\lambda = 64.8$) was selected to mitigate the risk of overfitting within the high-dimensional feature space.

The final model, refitted on the complete training set, achieved a test accuracy of 0.66 and a weighted F1-score of 0.66. As shown in the performance metrics (Figure 2.2a), the classifier demonstrated consistent reliability across the five clinical categories, with *Anxiety* and *Schizophrenia* emerging as the classes with the highest F1-scores (0.70 and 0.69, respectively). The Multiclass ROC curves (Figure 2.2b) yield AUC values between 0.86 and 0.91, confirming a strong discriminative capability.

Finally, the confusion matrix (Figure 2.2c) illustrates the model's balanced performance. The high consistency between the cross-validation score (0.65) and the test accuracy (0.66) indicates that the XGBoost model generalizes exceptionally well. The use of a shallow tree depth and robust regularization ensured that the model captured structural linguistic patterns rather than noise, successfully maintaining stable performance across unseen data.



(a) Performance metrics per class (Precision, Recall, F1-Score).

(b) Multiclass ROC curves.

(c) Confusion matrix.

Figure 2.2: XGBoost Evaluation

2.3 Multi-Layer Perceptron (MLP)

Complementing the ensemble methods, a deep learning architecture based on a Multi-Layer Perceptron (MLP) was implemented using the PyTorch framework. This approach was selected to leverage the representational power of neural networks in capturing high-order non-linear interactions within the 805-dimensional feature space. The input vectors were standardized using Z-score normalization (StandardScaler). This step was essential to place all features on a common scale, ensuring the optimization process could converge efficiently.

The network architecture features a "bottleneck" design intended to compress input features into abstract representations. It comprises fully connected layers paired with Batch Normalization to maintain training stability, ReLU functions to model complex non-linear patterns, and Dropout layers to ensure the model learns robust features rather than memorizing the training data. The training was hardware-accelerated using NVIDIA CUDA architecture.

Hyperparameter optimization was conducted via a systematic Grid Search, exploring combinations of learning rates, hidden layer dimensions, and dropout probabilities. The optimization identified the best configuration with a learning rate of 0.0005, a hidden dimension of 256, and a dropout rate of 0.5. The training

utilized the AdamW optimizer for decoupled weight decay and employed an Early Stopping mechanism (patience = 5) to automatically restore the best model weights when validation loss ceased to improve.

The training dynamics for the top-performing configurations are illustrated in Figure 2.3. The Loss curves (Left) demonstrate a stable convergence, where the validation loss (solid line) stabilizes around epoch 25, effectively triggering the Early Stopping mechanism to halt training before overfitting occurs. Correspondingly, the Accuracy trajectories (Right) confirm the model's generalization capability, with the validation accuracy reaching a plateau of approximately 69% despite the continued optimization on the training set.

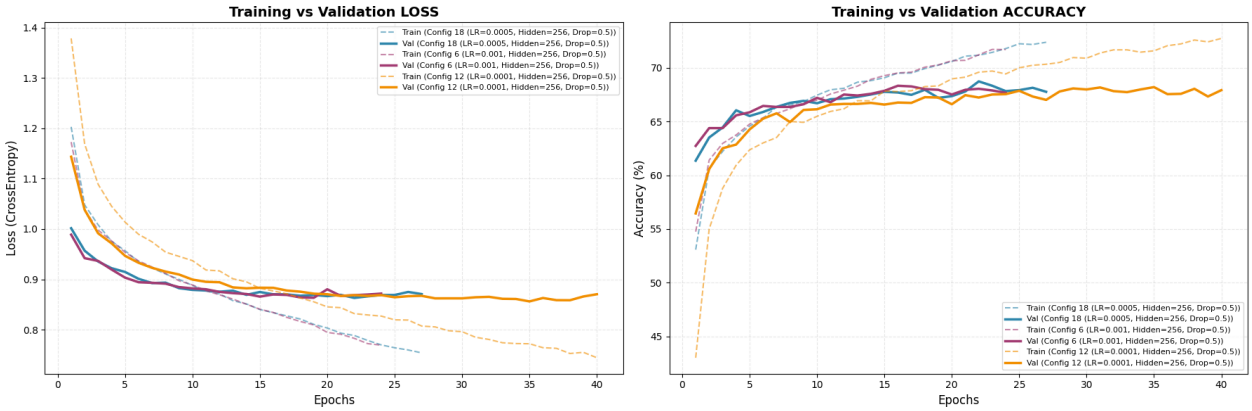
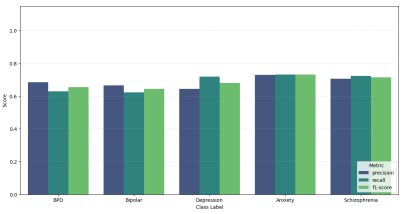


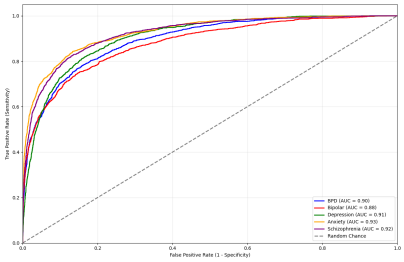
Figure 2.3: Training and Validation Dynamics. The plots compare Loss (Left) and Accuracy (Right) across different grid-search configurations.

The final model achieved an overall accuracy of 0.69 and a weighted F1-score of 0.68. As illustrated in the performance metrics (Figure 2.4a), the classifier shows strong performance across all categories, with Anxiety and Schizophrenia achieving the highest F1-scores. The Multiclass ROC analysis (Figure 2.4b), derived from the model's predicted probabilities, yields AUC values ranging from 0.88 to 0.93, validating the model's high probabilistic confidence.

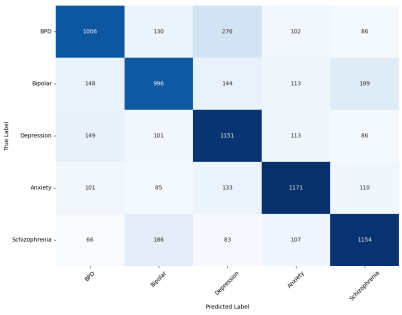
Finally, the confusion matrix (Figure 2.4c) demonstrates the network's classification behavior. The strong diagonal density confirms that the MLP successfully generalized the learned patterns to unseen mental health discourse without overfitting, matching the reliability of the ensemble approaches.



(a) Performance metrics per class (Precision, Recall, F1-Score).



(b) Multiclass ROC Curves.



(c) Confusion Matrix.

Figure 2.4: MLP Evaluation

3 Explainability

To ensure the reliability of the Multi-Layer Perceptron (MLP) and verify that its predictions are based on meaningful linguistic patterns rather than statistical artifacts, a post-hoc interpretability analysis was conducted. Since deep neural networks operate as "black boxes" with non-linear decision boundaries, understanding the specific contribution of each feature is essential for clinical validity. Consequently, the analysis

employed two complementary techniques: LIME, to examine the reasoning behind individual predictions, and SHAP, to provide a global overview of feature importance across the entire dataset.

3.1 LIME

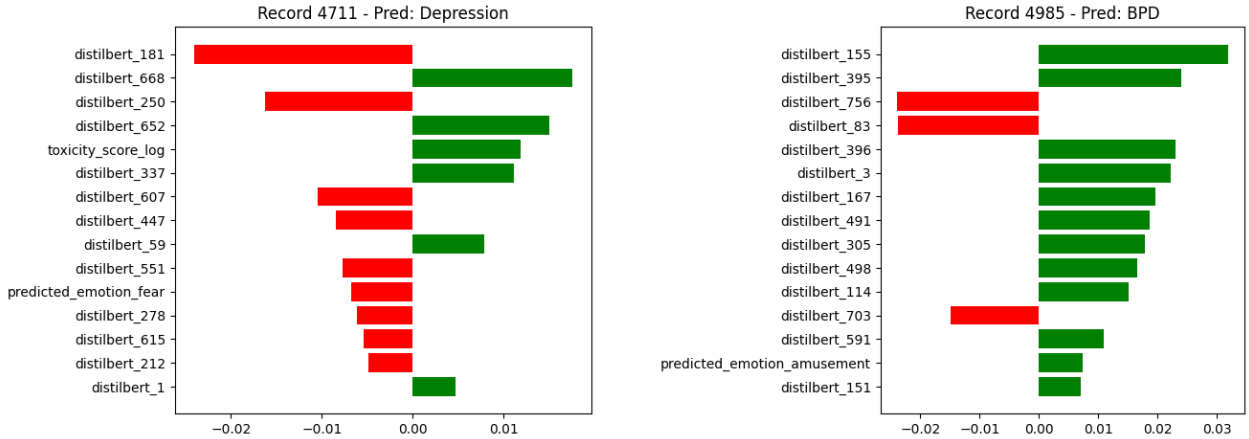
To interpret neural network's decisions, LIME (Local Interpretable Model-agnostic Explanations) was employed. Rather than attempting to explain the entire complex model at once, this technique analyzes individual predictions to reveal exactly which input features tipped the scale toward a specific diagnosis.

For this implementation, the analysis was conducted on a random subset of 50 samples from the test set. A consistent pattern emerged from these examples: the model demonstrates a "hybrid" reasoning process. While it primarily relies on the deep semantic context provided by the DistilBERT embeddings to understand the general meaning of the text, specific manual linguistic features such as toxicity levels or emotional markers, often act as the decisive factors that determine the final classification.

To demonstrate this interaction, we selected two illustrative records where the textual content clearly aligns with the model's predictive reasoning.

Analysis of Record 4711 (Predicted: Depression) As shown in Figure 3.1a, for Record 4711, the model correctly identified the text as *Depression*. Beyond the general tone of hopelessness, the analysis shows that the decision was heavily influenced by the high "toxicity" score. This fits perfectly with the user's angry frustration ("fucked by life"). Interestingly, the model also noted a lack of fear-related words, which helped it rule out Anxiety and settle on a diagnosis of depression.

Analysis of Record 4985 (Predicted: BPD) In Figure 3.1b, we see Record 4985, which describes a rapid mood shift ("top of the world" vs "in tears"). The model correctly flagged this as *BPD*, driven largely by the conflicting emotions found in such a short message. Notably, the detection of **amusement**, linked to the euphoric phrase "top of the world", played a crucial role, as it sharply contrasted with the sadness expressed immediately after.



(a) Record 4711 (Pred: Depression)

(b) Record 4985 (Pred: BPD)

Figure 3.1: LIME Local Explanations for two illustrative test records. **(a) Record 4711:** *"Well my insurance cut out and I can't afford my meds... Just feel like I keep getting fucked by life..."* - The prediction is reinforced by the detected `toxicity_score_log`. **(b) Record 4985:** *"It's ridiculous. One minute you're on top of the world... and then the next you're in tears. Both are real."* - The model captures the explicit emotional volatility characteristic of BPD disorder.

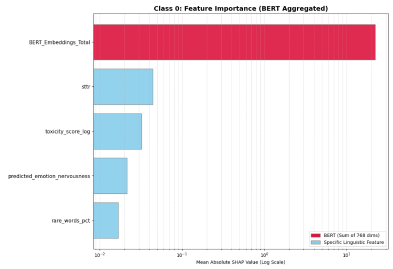
3.2 SHAP

While LIME focuses on individual predictions, SHAP (SHapley Additive exPlanations) was employed to understand the global decision-making criteria of the model across the entire dataset. To handle the computational complexity of the neural network, the `DeepExplainer` algorithm was used, summarizing the training data into 150 weighted centroids to approximate the background distribution.

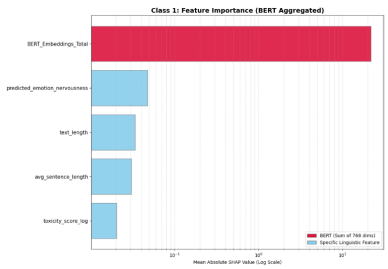
A key challenge in this analysis was comparing the 768 abstract dimensions of the DistilBERT embeddings against the 37 explicit manual features. To visualize this effectively, we applied an aggregation strategy: the SHAP values of all BERT dimensions were summed to represent the total "semantic content" (red bars in Figure 3.2), while manual features were kept separate (blue bars). Note that the plots utilize a logarithmic scale due to the massive cumulative contribution of the embedding layer.

As illustrated in Figure 3.2, the contextual embeddings consistently dominate the feature importance, confirming that the model primarily relies on the vector space to determine the context. However, the specific manual features that act as secondary discriminators vary significantly by class, revealing the model's linguistic sensitivity:

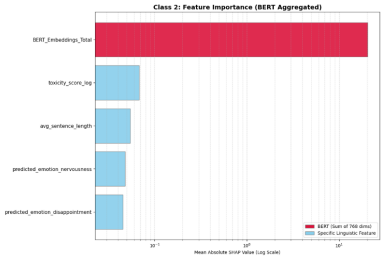
- **BPD (Class 0) and Schizophrenia (Class 4):** As shown in Figures 3.2a and 3.2e, the model places high importance on `sttr` (Standardized Type-Token Ratio), a metric of lexical diversity. This suggests that the complexity or repetitiveness of the vocabulary is a key marker for these conditions.
- **Bipolar (Class 1):** In Figure 3.2b, text length features (`text_length`, `avg_sentence_length`) emerge as top predictors, likely capturing the verbose nature of manic phases or the brevity of depressive episodes.
- **Depression (Class 2):** Figure 3.2c highlights `toxicity_score_log` as the strongest specific feature. The presence of hostile or self-deprecating language is a primary signal the model uses to isolate depression from other mood disorders.
- **Anxiety (Class 3):** As seen in Figure 3.2d, the model strongly leverages the `predicted_emotion_nervousness`, confirming that explicit expressions of worry and fear are the most reliable discriminators for anxiety.



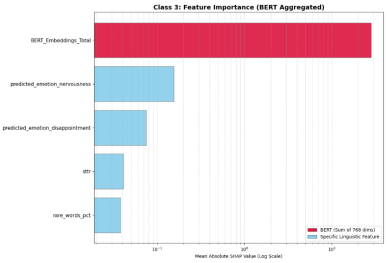
(a) Class 0 (BPD)



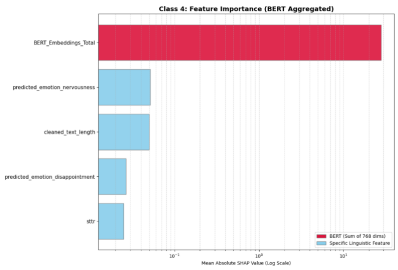
(b) Class 1 (Bipolar)



(c) Class 2 (Depression)



(d) Class 3 (Anxiety)



(e) Class 4 (Schizophrenia)

Figure 3.2: Aggregated SHAP Feature Importance per Class (Log Scale). The Red bars represent the cumulative contribution of DistilBERT embeddings, while Blue bars show the top specific linguistic features. The analysis reveals distinct signatures for each disorder, such as `toxicity` for Depression and `sttr` for BPD.

Conclusion

This study investigated the potential of NLP to identify mental health conditions from unstructured text, aiming to support the growing need for accessible support tools. A comprehensive pipeline was developed, combining the deep semantic understanding of DistilBERT with specific linguistic features. The data preparation phase proved particularly challenging; as the texts originated from online forums, their informal and conversational nature required rigorous preprocessing strategy.

Among the evaluated architectures, the Multi-Layer Perceptron (MLP) emerged as the most effective, achieving a test accuracy of 0.69. However, it is important to note that all three classifiers proved effective for the task; the SVM and XGBoost models also demonstrated robust capabilities, reaching accuracy levels only marginally lower than the neural network.

Furthermore, the application of LIME and SHAP enabled an in-depth examination of the neural network's "black box". These interpretability techniques revealed a hybrid reasoning logic: the model relies on the general context while simultaneously identifying specific linguistic cues such as toxic language or emotional shifts, to determine the final classification.

Finally, it is crucial to note that this project does not aim to replace medical professionals or clinical diagnoses. Instead, it serves as a foundational step to demonstrate the diagnostic value of language analysis. By illustrating how specific lexical patterns reflect psychological states, this work highlights the importance of comprehending the nuances of written text, paving the way for future supportive tools designed to enhance, rather than substitute, human expertise in mental healthcare.

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